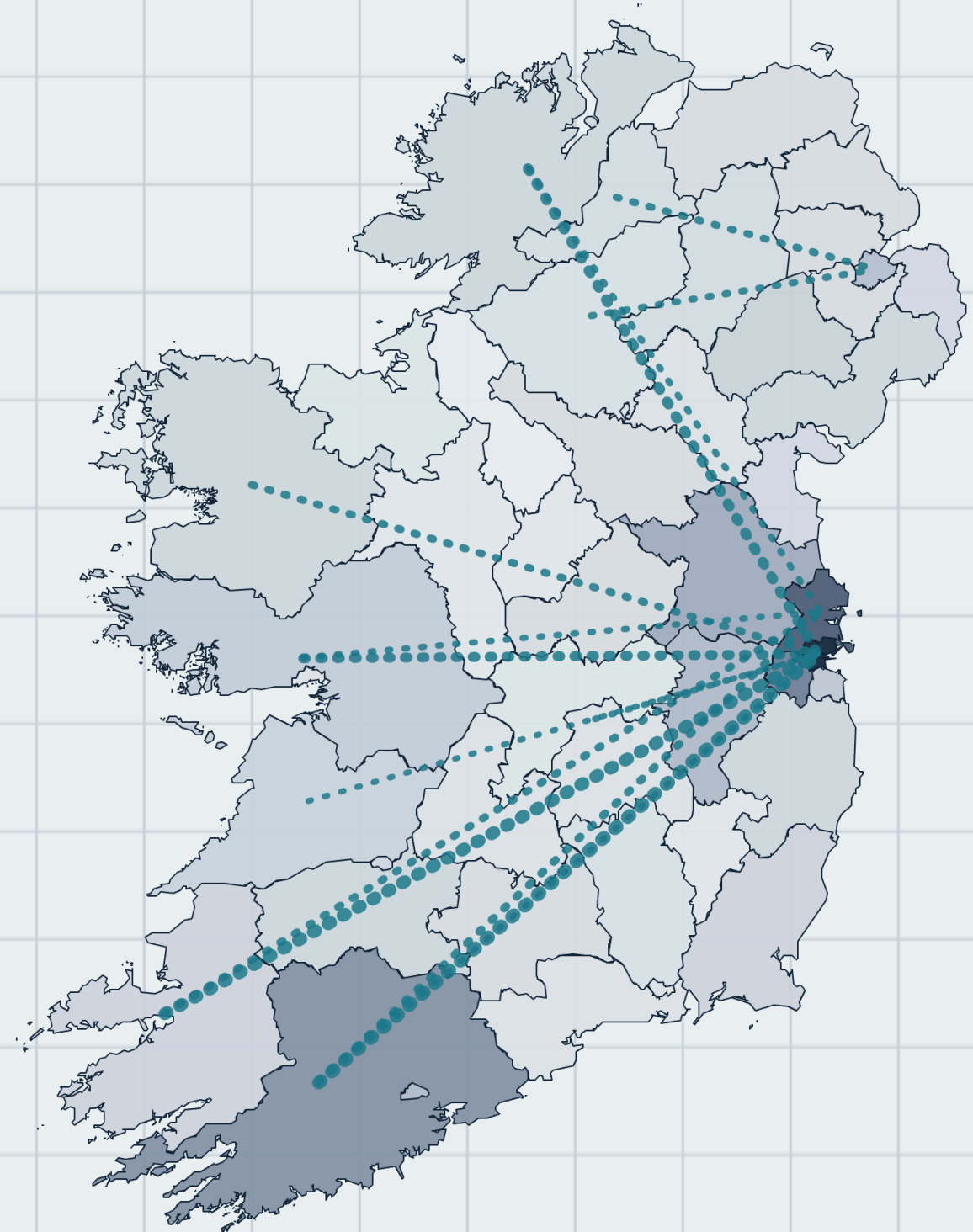


HACKATHON 2026 • EIRGRID + CSO OPEN DATA

GridSense

Forecast where Ireland's load will peak.
Route renewables there first.



SECTION 01 / 07

The problem

Wind gets switched off while the grid says no.

The problem

What GridSense does

Who it's for

Impact

Technical details

Live demo

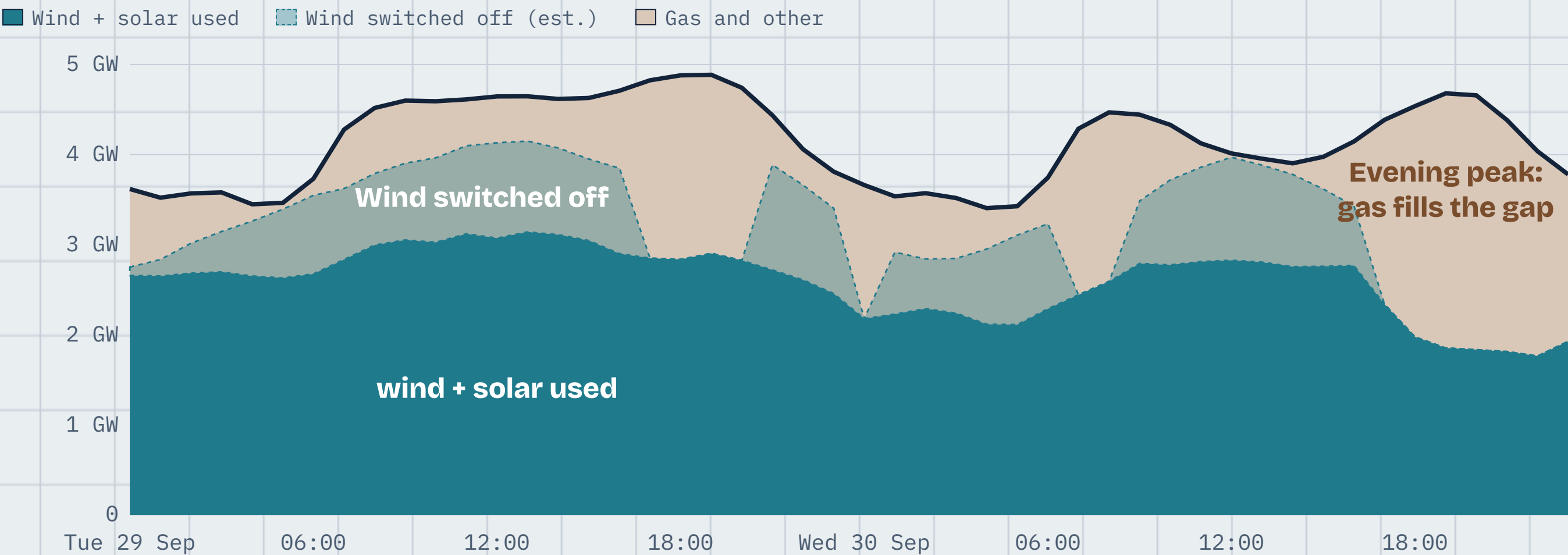
What's next

THE PROBLEM

Wind gets switched off. Then gas runs the peak.

37.6%

island wind + solar share, 12 months

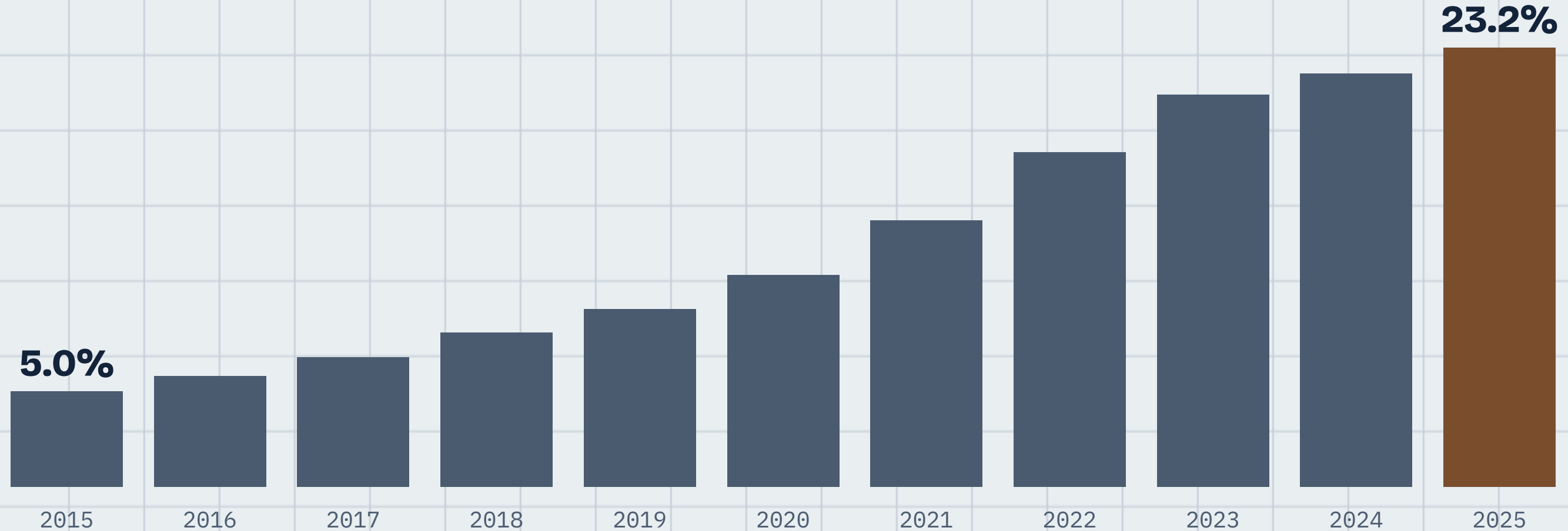


EirGrid Smart Grid Dashboard, ROI hourly, 29-30 Sep 2026. Switched-off wind estimated as wind forecast - actual when the grid is $\geq 60\%$ renewable.

THE PROBLEM

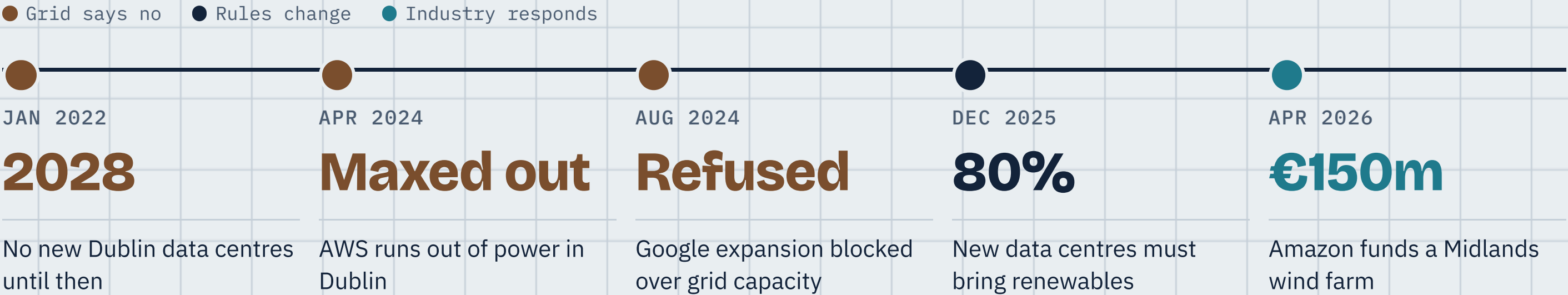
Data centres now use **23.2%** of the Republic's electricity.

6.2×
more than in 2015



CSO MEC02: data-centre share of ROI metered electricity, by year, 2015–2025.

The grid said no. The rules changed.



Where GridSense fits: hour-by-hour, per-area proof of renewable use.

Sources: RTÉ / DCD, Jan 2022 • The Register via DCD, 10 Apr 2024 • Techerati, Sep 2026 • CRU decision via William Fry, 15 Dec 2025 • Irish Examiner, 27 Apr 2026

SECTION 02 / 07

What GridSense does

Forecast the peak. Route wind to it first.

The problem

What GridSense does

Who it's for

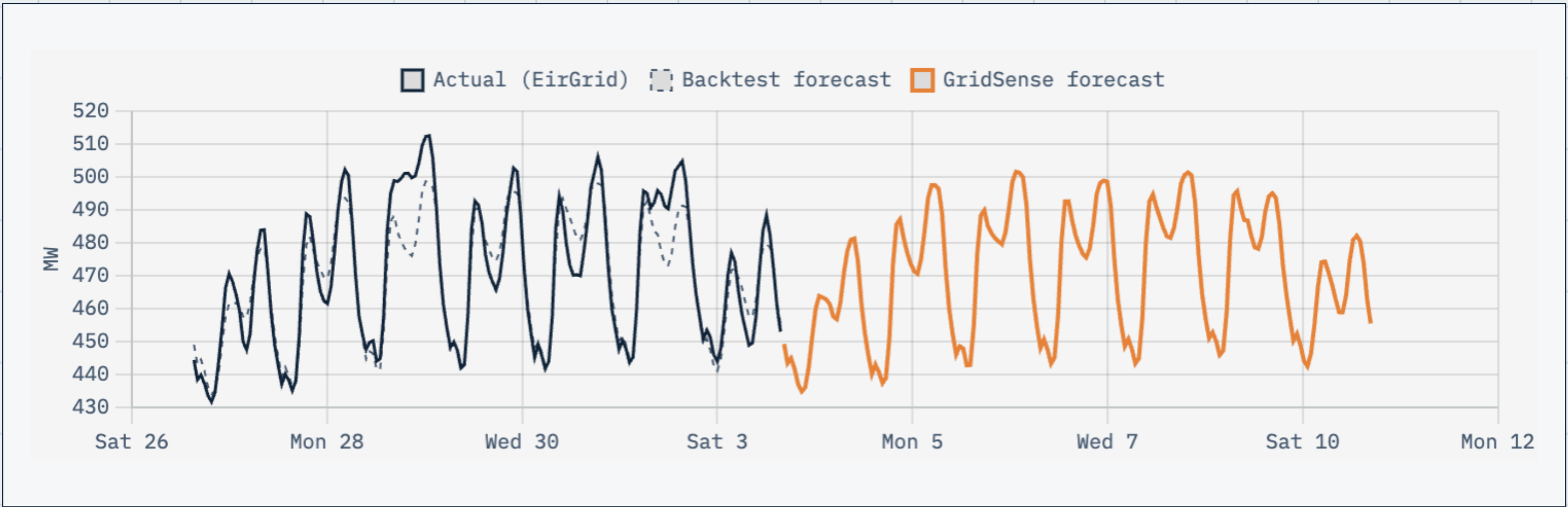
Impact

Technical details

Live demo

What's next

We forecast every area's peak, a week ahead.



2.8%

backtest error, ROI

45

areas, hourly, 7 days

696 MW

Dublin City, Tue 18:00

Then we book spare wind for the biggest peak.

Cork County → Dublin City

485 MWh

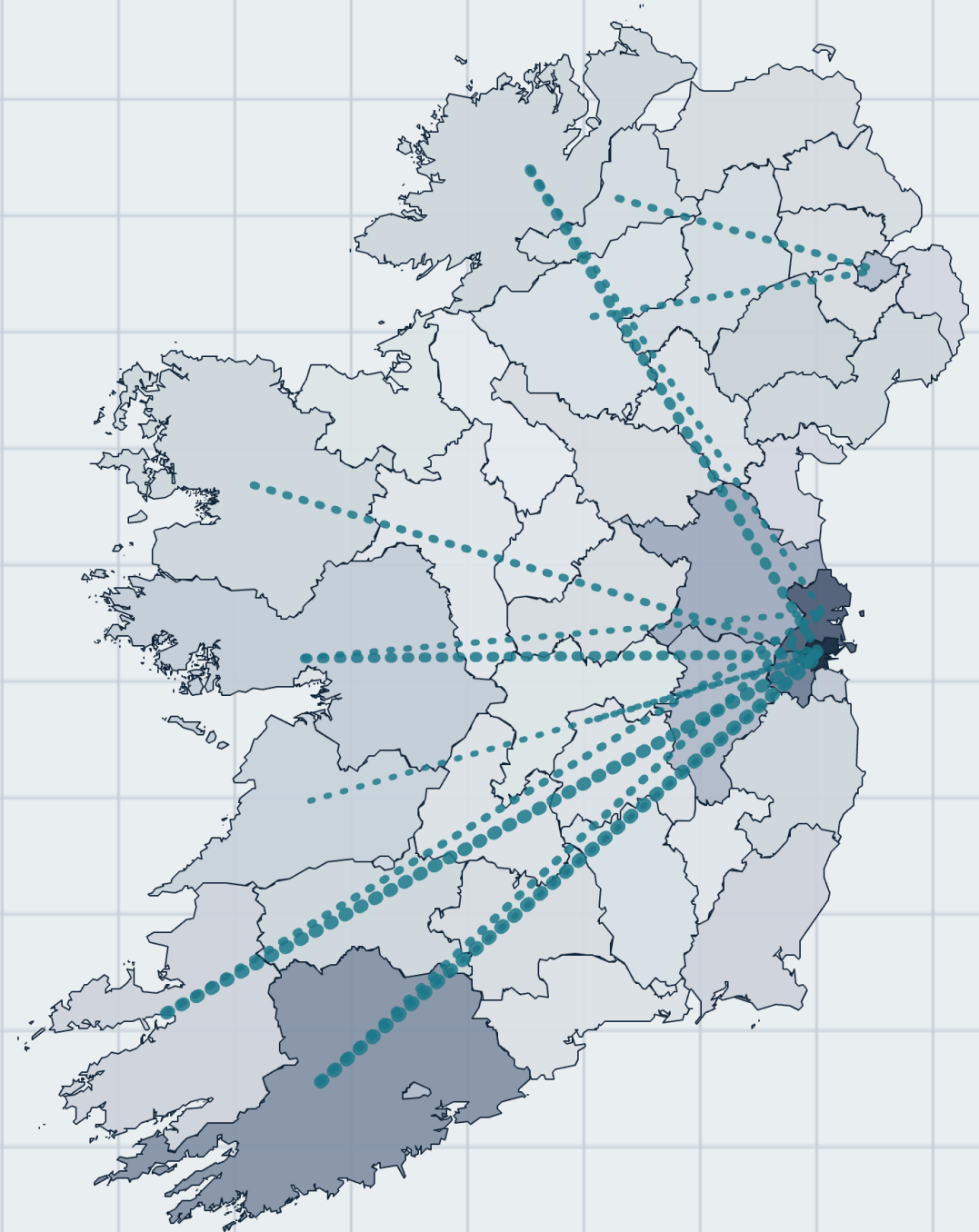
Kerry → Dublin City

461 MWh

Donegal → Dublin City

364 MWh

12.8 GWh planned this week



And we prove the cut, area by area.

-9.3 kt CO₂

AVOIDED WITH GRIDSENSE • 12 MONTHS

~~670.3 kt~~ → **661.0 kt** **-1.4%**

Fingal gets **82.3 GWh** more wind, delivered in its forecast peak hours, when the grid would otherwise burn gas. Wind + solar share goes from **37.6%** to **39.6%**. The load stays the same and nothing changes on site.

Fingal, Ireland's data-centre hub • 12 months • live dashboard

SECTION 03 / 07

Who it's for

Data centres, government, new industry,
households.

The problem

What GridSense does

Who it's for

Impact

Technical details

Live demo

What's next

Same workload. More wind behind it.

1,845 MW

forecast peak, this week

+299 GWh

recovered wind at peak hours

-33.8 kt

CO₂ cut, from 2.36 Mt

What a data centre gets.

24/7

Hourly-matched carbon-free evidence

Scope 2

Location-based figures, per site

80%

An answer to the CRU rule

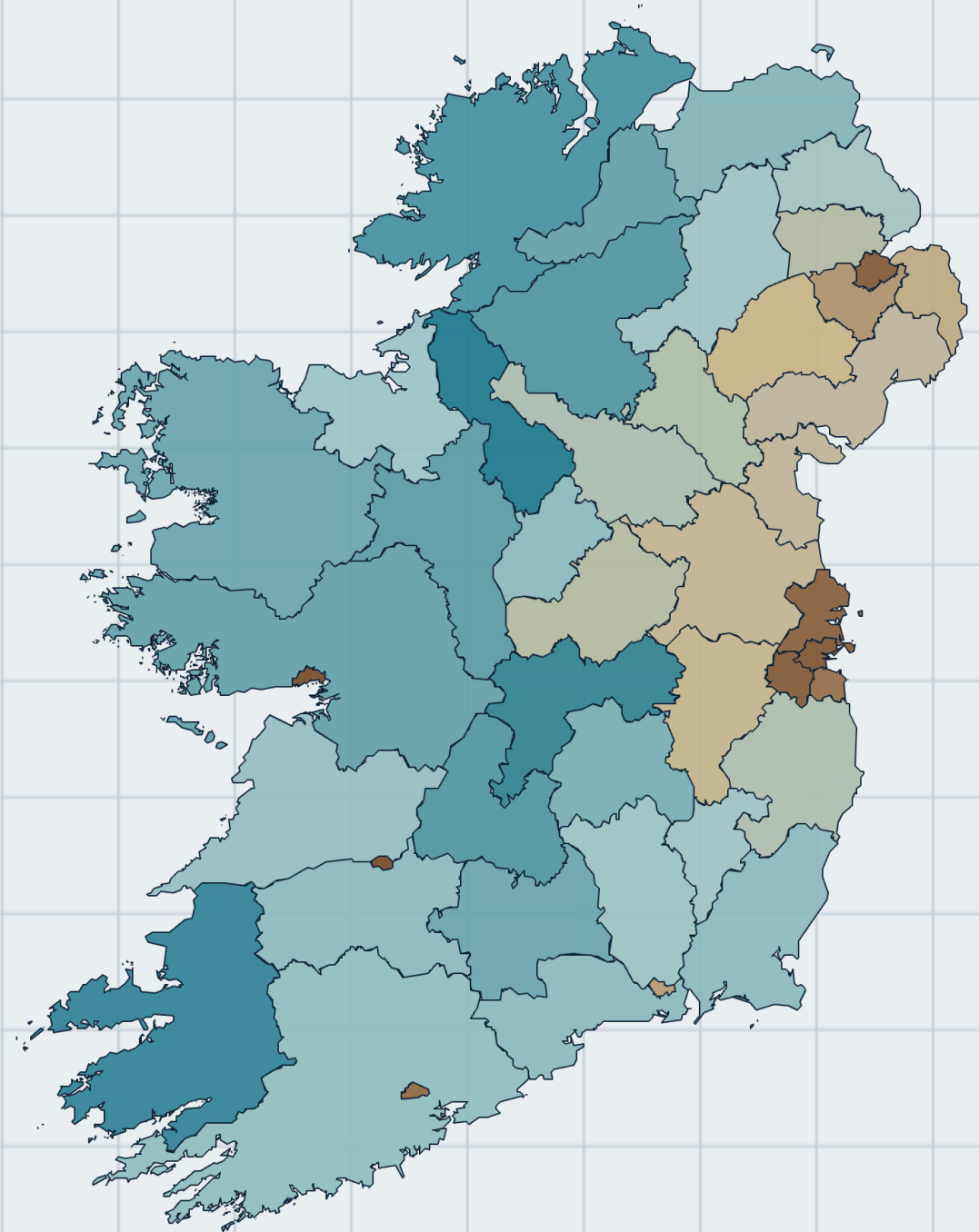
No change to workloads or SLAs.

The west makes the wind. Dublin uses it.

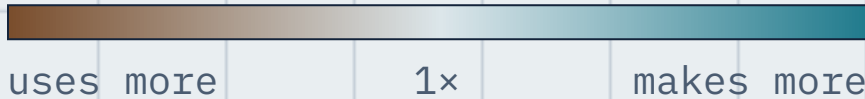
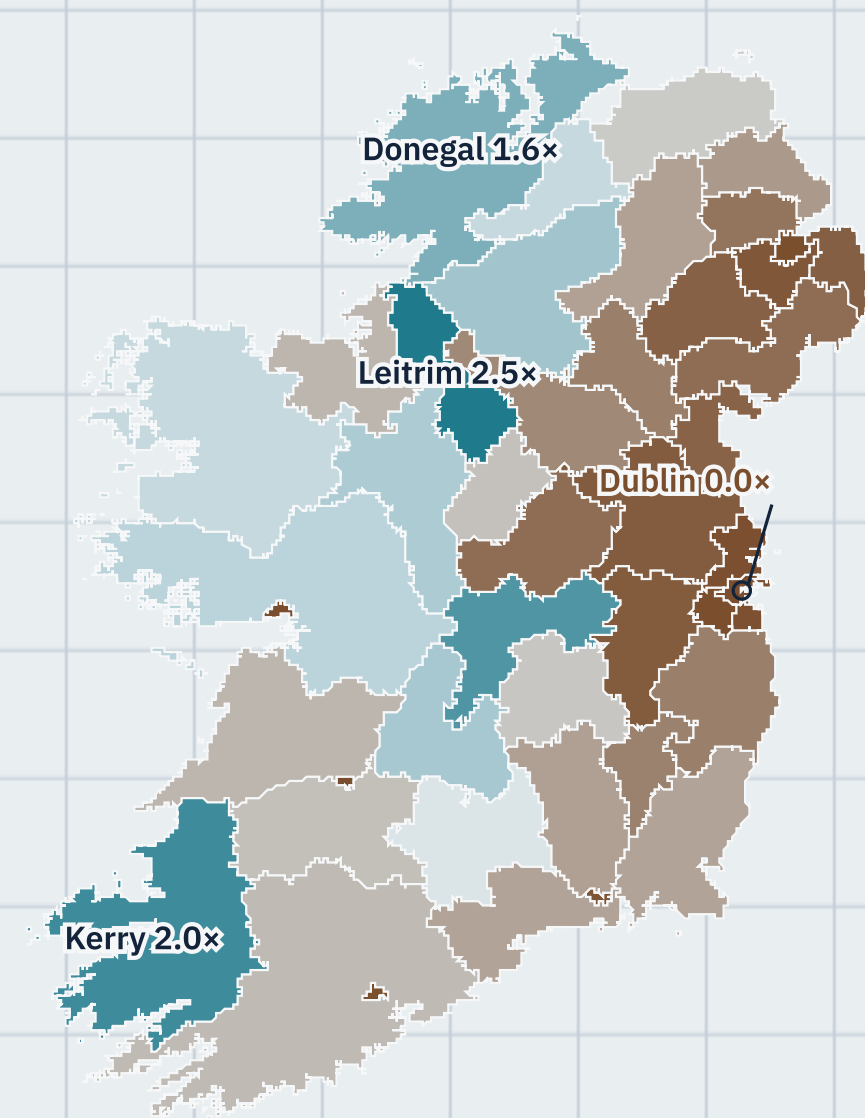
 **Leitrim 1.9× • Kerry 1.6×**
generate more than they use

 **Dublin ~0.01×**
imports almost all of it

Local generation ÷ use, last 3 months. Where the next
wind farm and cable should go.



Build where the wind is.



2.5x

Leitrim makes 2.5x the renewables it uses

0.09x

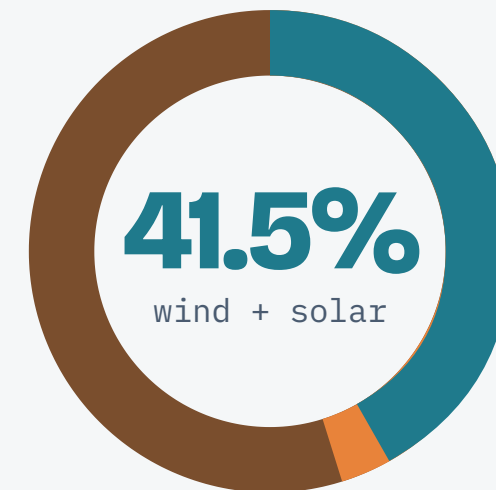
Meath and Kildare: the data-centre corridor



Site near the surplus, not the congestion

"How much of my
power was
renewable?"

DUBLIN CITY • LAST MONTH



44.8% with GridSense

120 kg CO₂ per resident

SECTION 04 / 07

Impact

Twelve months of real EirGrid data.

The problem

What GridSense does

Who it's for

Impact

Technical details

Live demo

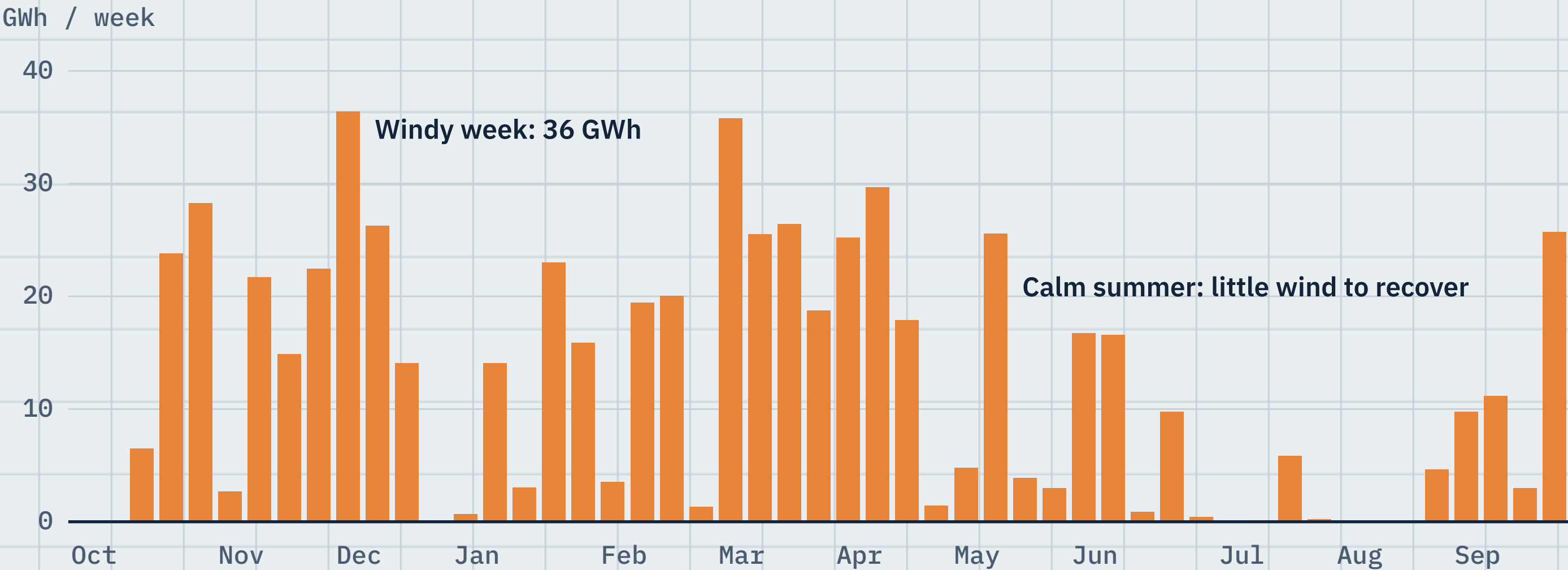
What's next

What a year of GridSense does.



Modelled on real EirGrid data, Oct 2025 – Oct 2026. Same load, same weather.

Windy weeks do the heavy lifting.



SECTION 05 / 07

Technical details

The data, the forecast, the routing.

The problem

What GridSense does

Who it's for

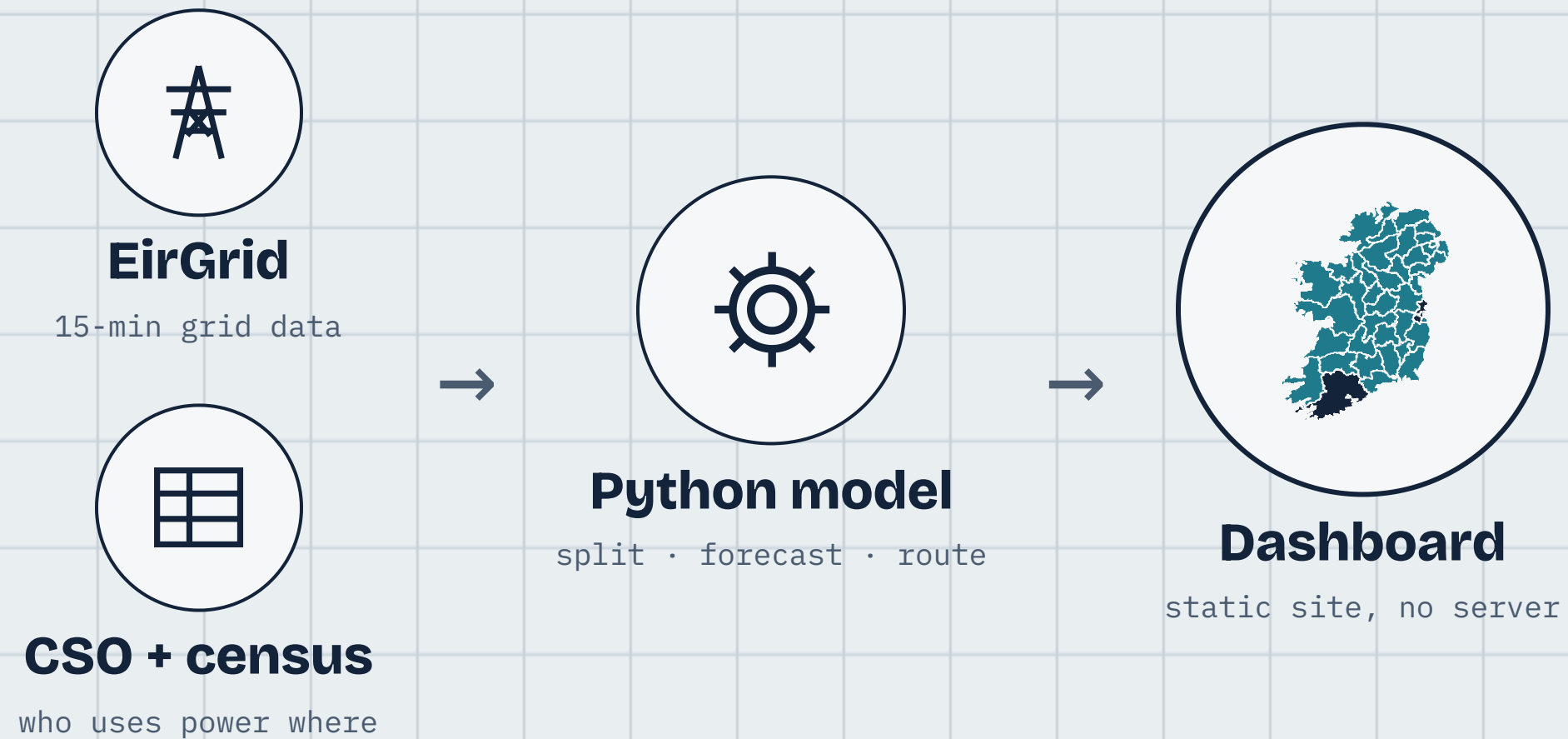
Impact

Technical details

Live demo

What's next

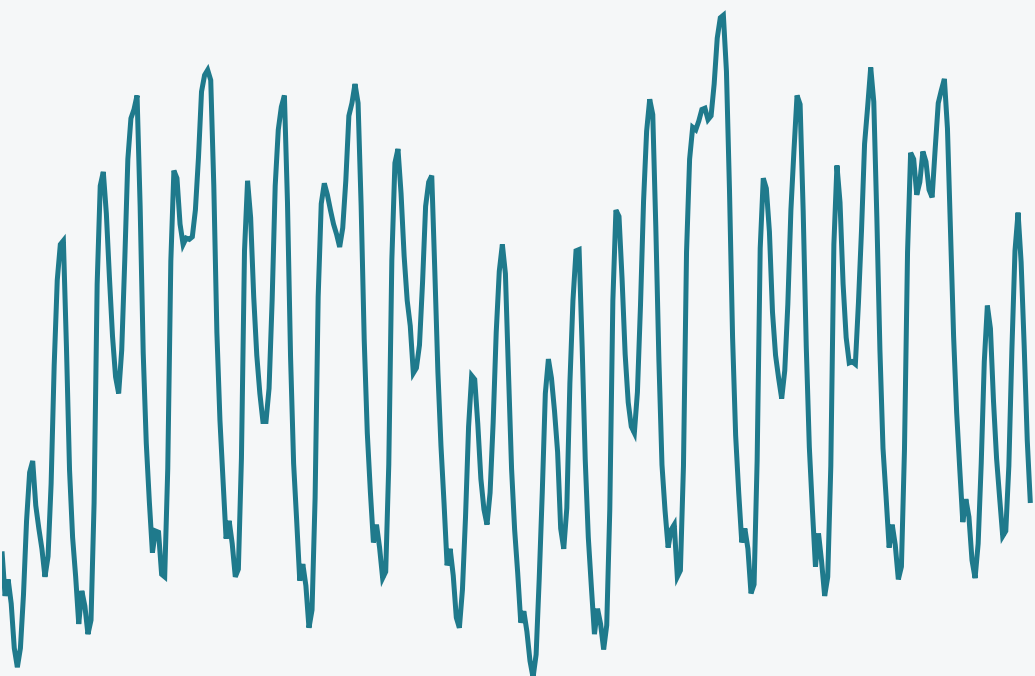
Open data in. One dashboard out.



All public. All real.

EirGrid

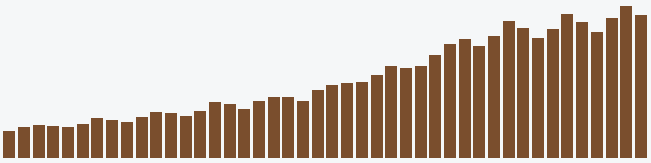
demand · wind · solar · CO₂ · 15-min



ROI demand, last 14 days

CSO MEC02

data centres: 23.2% of ROI
power



2015Q1 → 2025Q4

NISRA

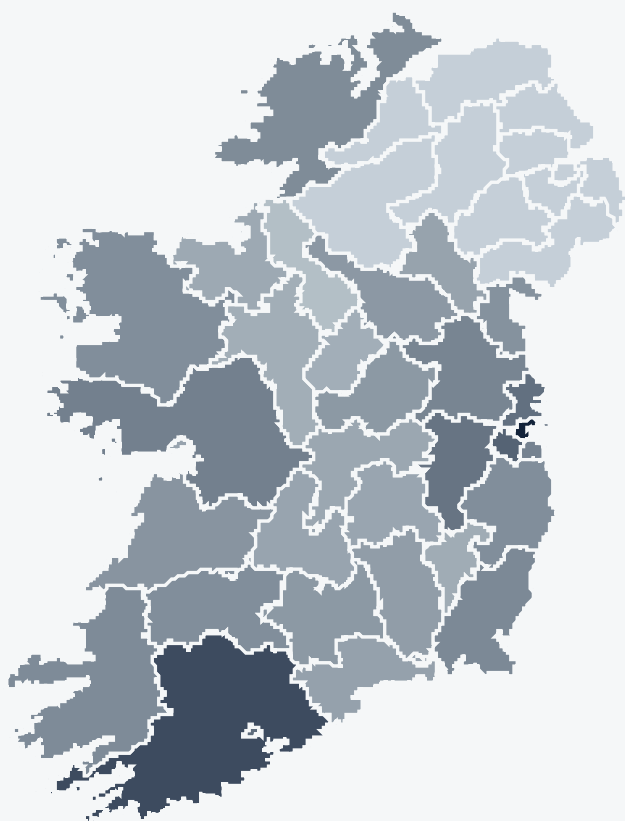
NI
population

OSi · OSNI

boundaries

CSO MEC03

use by county

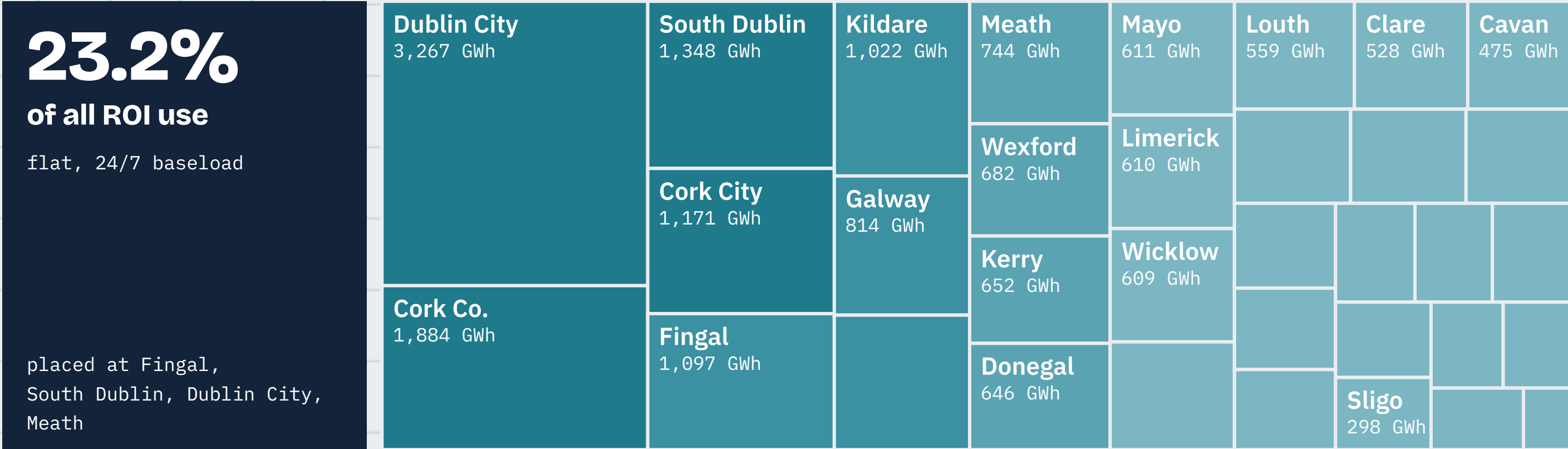


Five steps from open data to a routing plan.



Step 2: share the national load out to 45 areas.

CSO MEC02 · DATA CENTRES CSO MEC03 · METERED USE PER COUNTY + DUBLIN POSTCODE, 2025

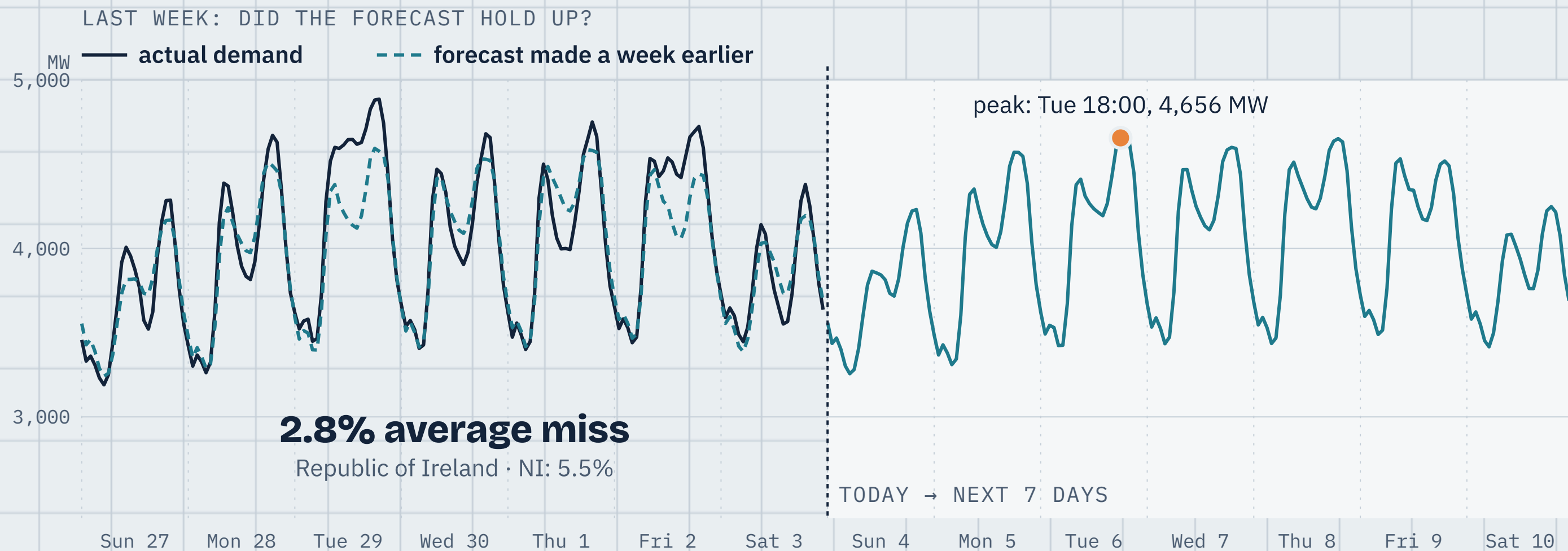


tile size = electricity really used

+ NI: 11 districts, by population

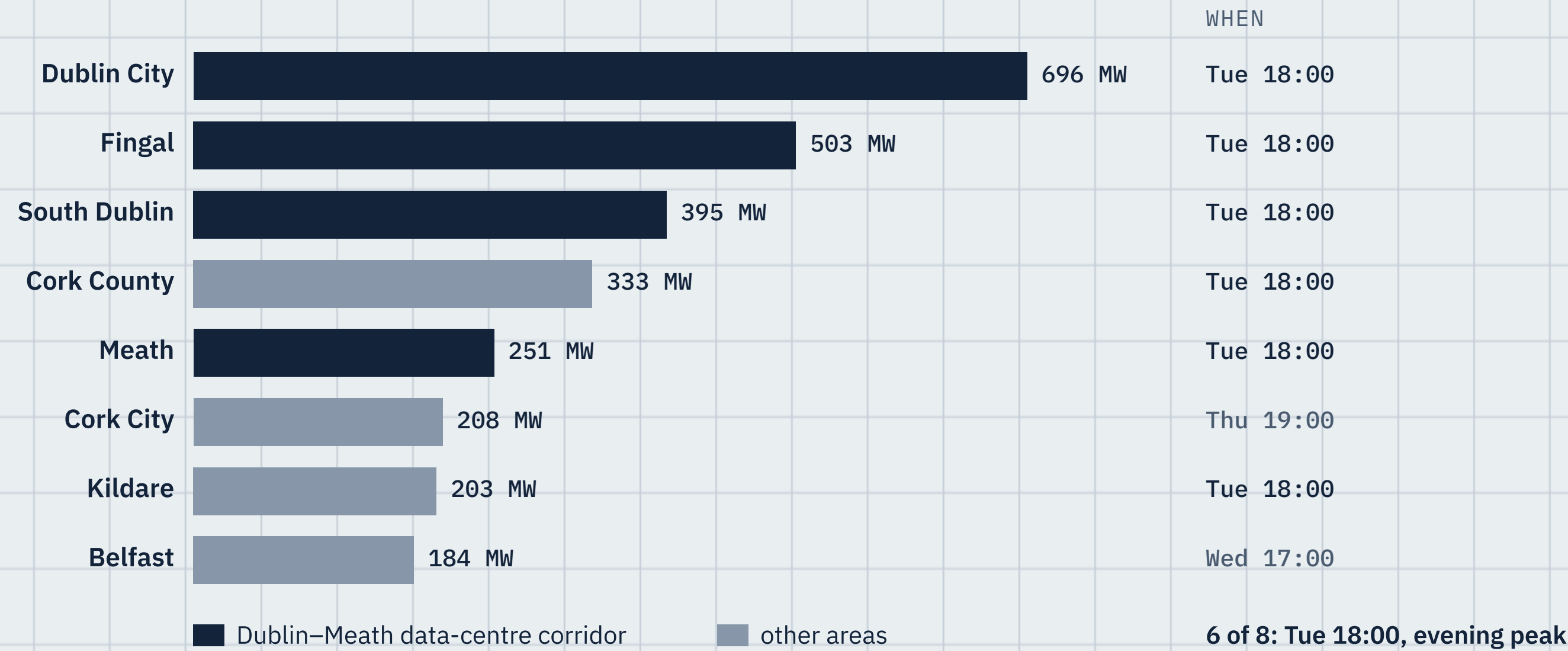
✓ adds up to EirGrid demand, ±1%

Step 3: forecast every hour of next week.



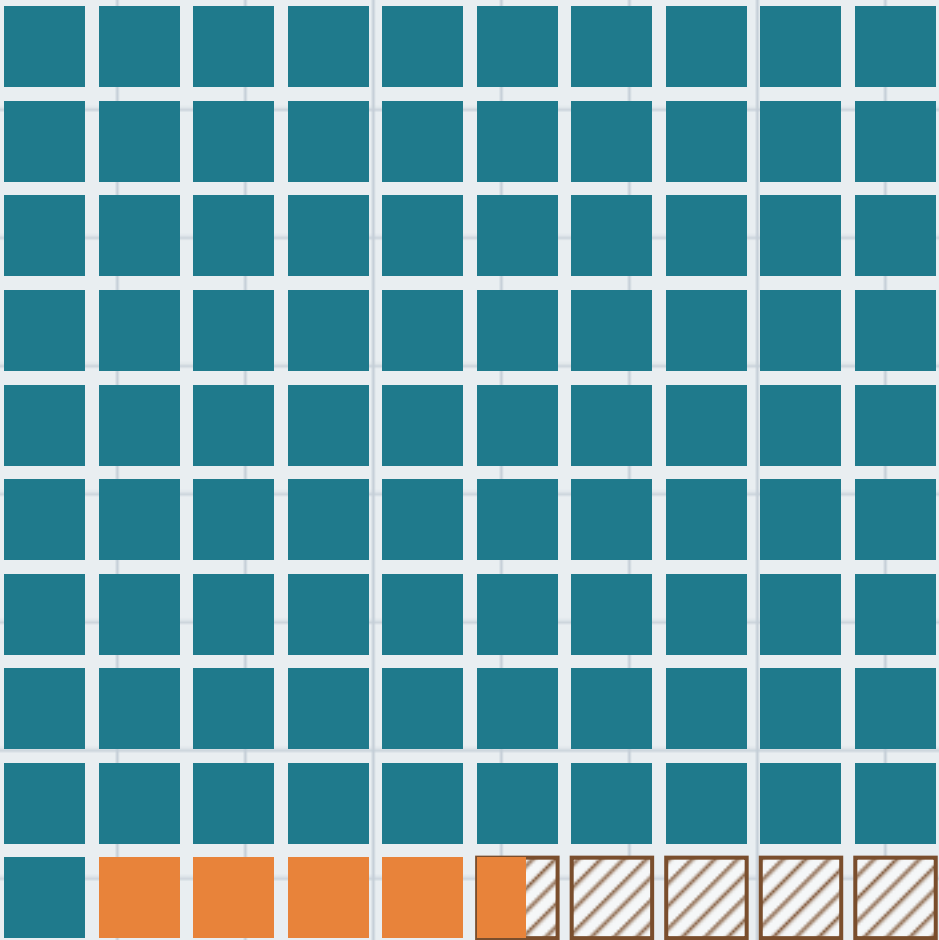
how: average of the same hour over the last 8 weeks × last week's trend

Step 4: rank where next week's peaks hit.



The biggest peaks get first call on spare wind →

Step 5a: find the wind that gets switched off.



every square = 1% of ROI wind, 12 months

91% reaches customers

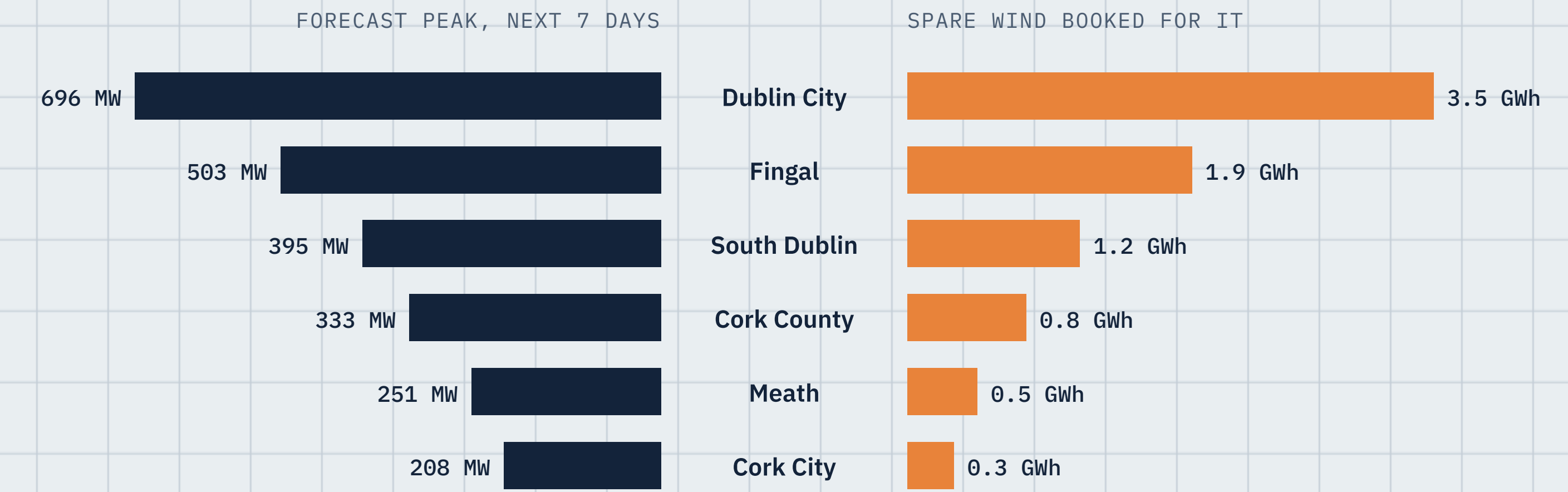
wind that the grid could absorb

9.2% dispatched down

wind farms told to stop: forecast – actual, when the grid is $\geq 60\%$ renewable
EirGrid reports $\sim 10\%$ ✓

Half of it ($\approx 4.6\%$) is grid congestion.
GridSense schedules that part into forecast peaks.

Step 5b: share it out, biggest peaks first.



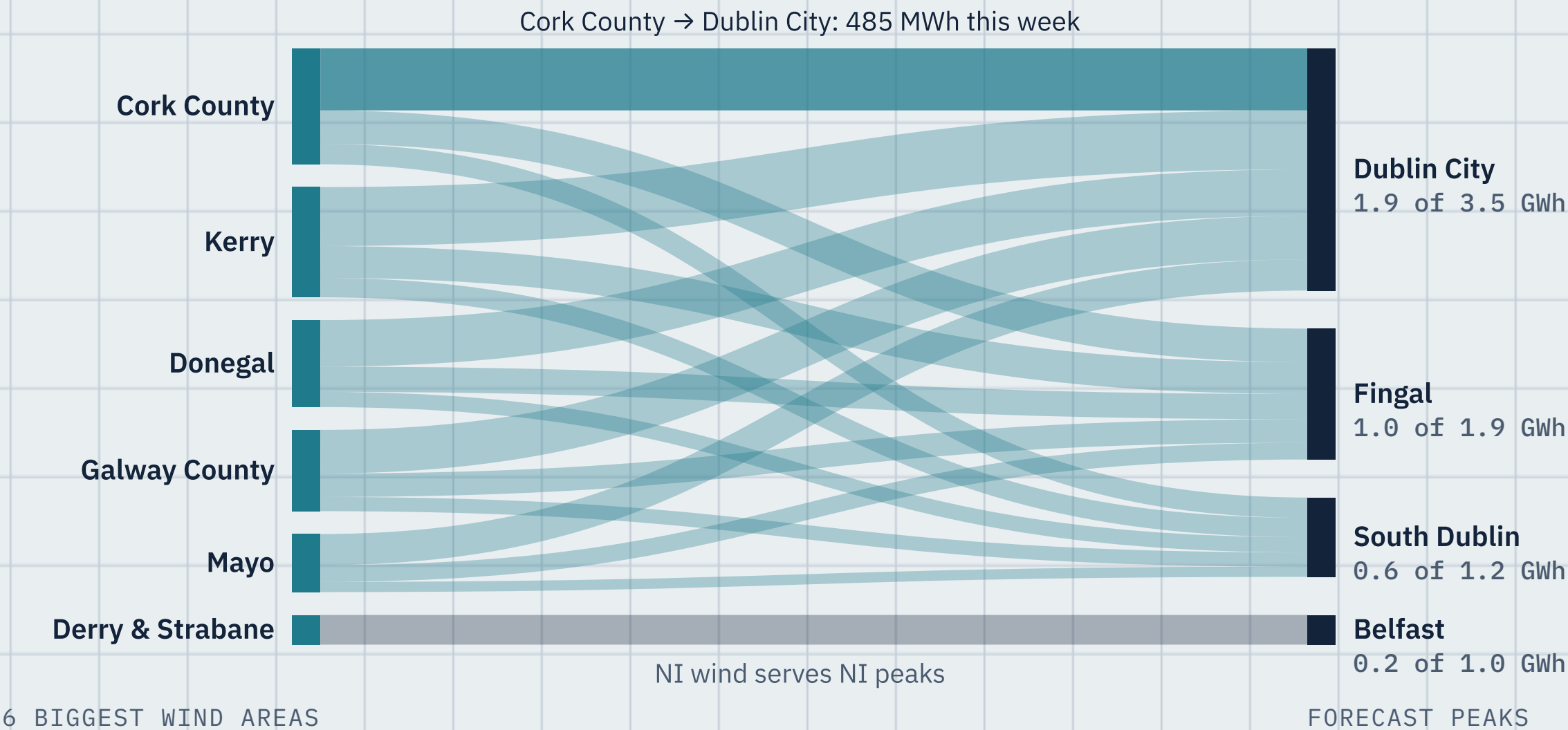
Share grows with peak²: Dublin City's peak is 1.4× Fingal's, so it gets 1.8× the wind. Everyone gets some.

into each day's 6 highest forecast hours

capped at the area's gas-fired load in those hours

leftover re-shared

Step 5c: every flow has a source and a number.

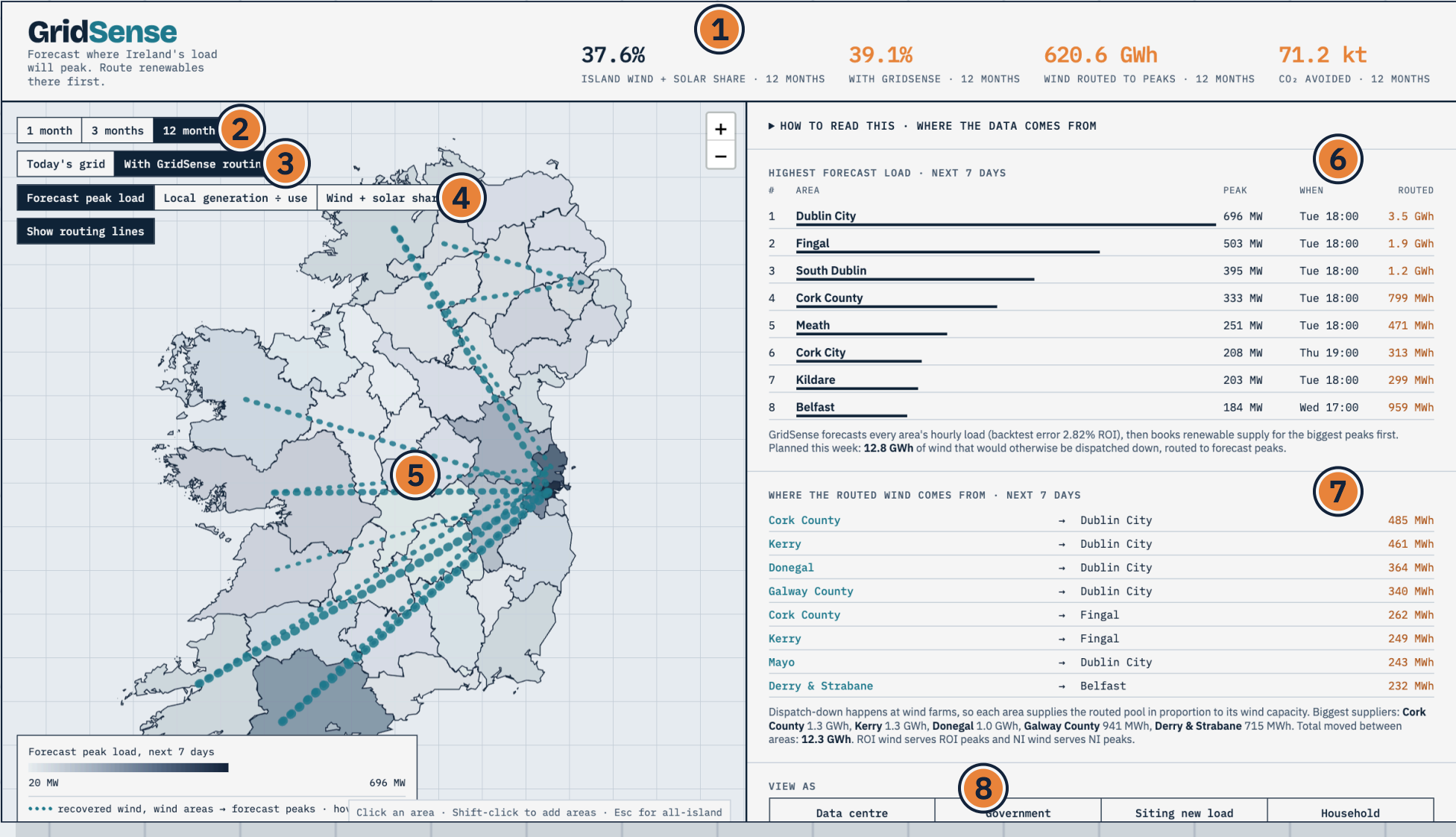


thicker = more energy, next 7 days

each wind area supplies in line with its wind capacity

12.8 GWh planned

One screen: where peaks hit, where wind comes from.



- 1 Island headline numbers
- 2 Period: 1, 3 or 12 months
- 3 Today vs with GridSense
- 4 Three map colourings
- 5 Routing lines: wind → peaks
- 6 Biggest peaks, next 7 days
- 7 Where the wind comes from
- 8 View as: data centre, government...

Click any area: see what GridSense saves there.

SELECTED AREA

Fingal

2025-10-02 → 2026-10-03 · Republic of Ireland

37.6%

→

39.6%

1

wind + solar share today → with GridSense routing

-9.3 kt CO₂

2

AVOIDED WITH GRIDSENSE · 12 MONTHS

670.3 kt

→

661.0 kt

-1.4%

3

Fingal gets **82.3 GWh** more wind, delivered in its forecast peak hours, when the grid would otherwise burn gas. Wind + solar share goes from **37.6%** to **39.6%**. The load stays the same and nothing changes on site.

4.11 TWh

electricity used

670.3 kt

CO₂ footprint today

503 MW

4

forecast peak, next 7 days

- 1 Wind + solar share: today → with GridSense
- 2 CO₂ avoided, this period
- 3 Footprint before → after
- 4 Next week's forecast peak

What we tried, and what we changed.

Allocation · Fingal's share

biggest peak takes all

0 GWh

share grows with peak²

1.9 GWh / week

Greedy starved the #2 peak. Water-filling fixed it.

Product · 12 months

shift data-centre load

353 GWh · 48.6 kt CO₂

forecast-led routing

621 GWh · 71.2 kt CO₂

Routing wins, and customers change nothing.

Area split · forecast peak

by population

Dublin City 596 MW

Fingal 565 MW

by CSO metered use

Dublin City 696 MW

Fingal 503 MW

Real consumption moved the peaks.

Reality check · wind switched off

EirGrid reports

~10%

our estimate

9.2%

Our dispatch-down estimate matches EirGrid.

Every build checks itself. If one fails, it stops.

```
$ python3 build_data.py
```

```
CS0 MEC02: data centres = 23.2% of ROI metered electricity
```

```
ROI: forecast backtest MAPE 2.8% • NI: 5.5%
```

- ✓ areas add up to EirGrid system demand within 1%
- ✓ CS0 county split adds up to the CS0 total
- ✓ GridSense only adds renewables and only cuts CO₂
- ✓ no area goes past 100% renewable
- ✓ biggest peak gets at least as much as the 5th
- ✓ flows add up to what's routed within 5%

```
$ node chk.cjs → headless Chrome clicks every control: no JS errors, expected numbers
```

What's measured, what's modelled, what replaces it.

 Measured

- EirGrid demand, wind, solar, CO₂ · every 15 min
- EirGrid wind forecast
- CSO data-centre share · 23.2%
- CSO metered use per county
- Forecast error · 2.8% / 5.5%

 Modelled

- Each area's hourly shape → ESB Networks smart meters
- County wind + solar capacity → EirGrid / SONI generator lists
- Dispatch-down estimate → EirGrid dispatch-down reports
- 50% recoverable → live routing with EirGrid + SONI

What judges ask.

“You can't route electrons.”

| True. We route schedules: transmission, storage, allocation.

“Why is every area ~32.9%?”

| Generation is only published grid-wide. That's the gap we fill.

“Why such a simple forecast?”

| 2.8% error, fully explainable. Weather + ML next.

“Is 621 GWh realistic?”

| $13 \text{ TWh wind} \times \sim 9\% \text{ switched off} \times \text{half} \approx 600 \text{ GWh}.$

“Who pays?”

| Data centres (24/7 CFE, Scope 2), EirGrid, the CRU.

SECTION 06 / 07

Live demo

The real dashboard, all 45 areas.

The problem

What GridSense does

Who it's for

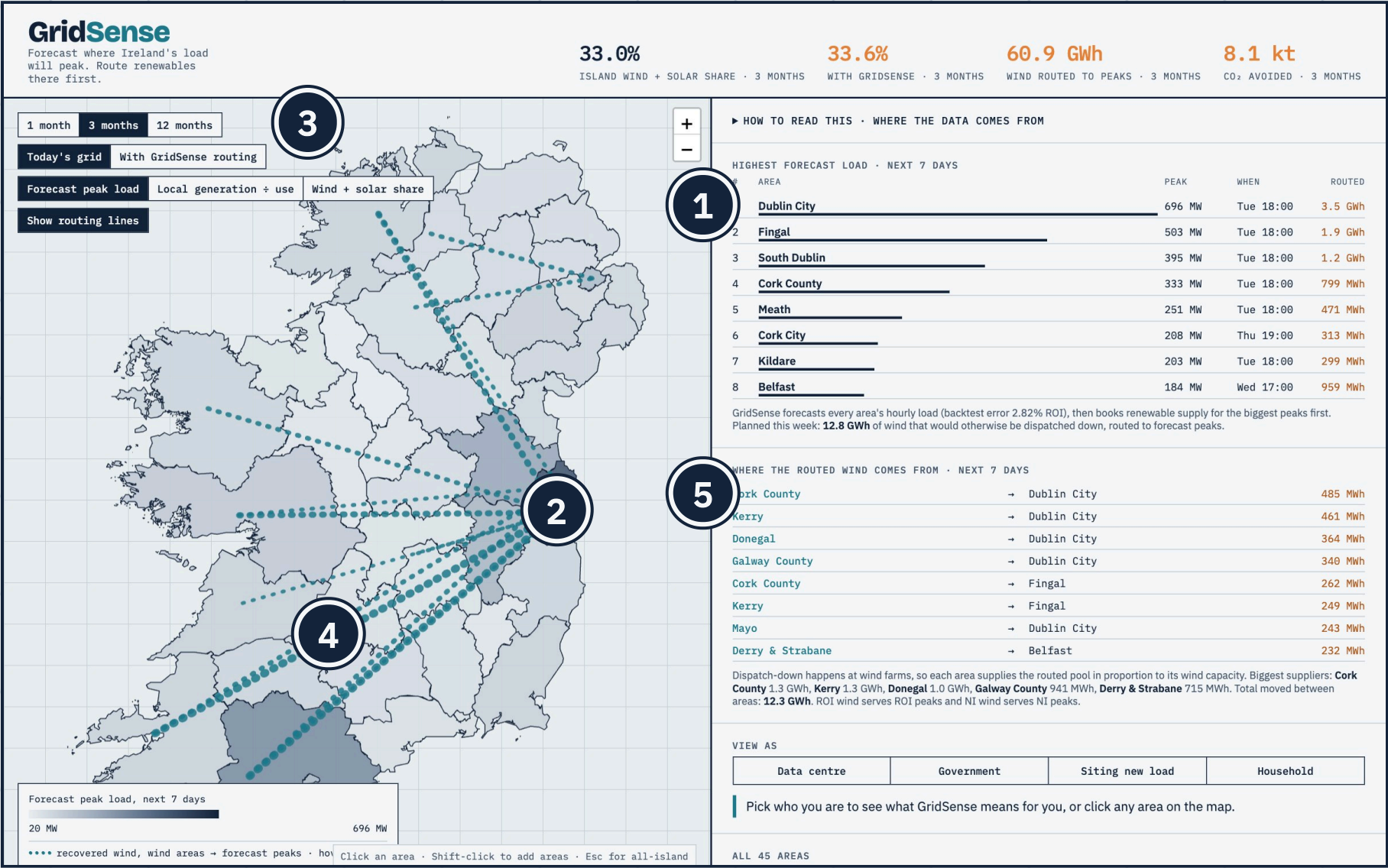
Impact

Technical details

Live demo

What's next

Follow the wind.



SECTION 07 / 07

What's next

From prototype to live routing.

The problem

What GridSense does

Who it's for

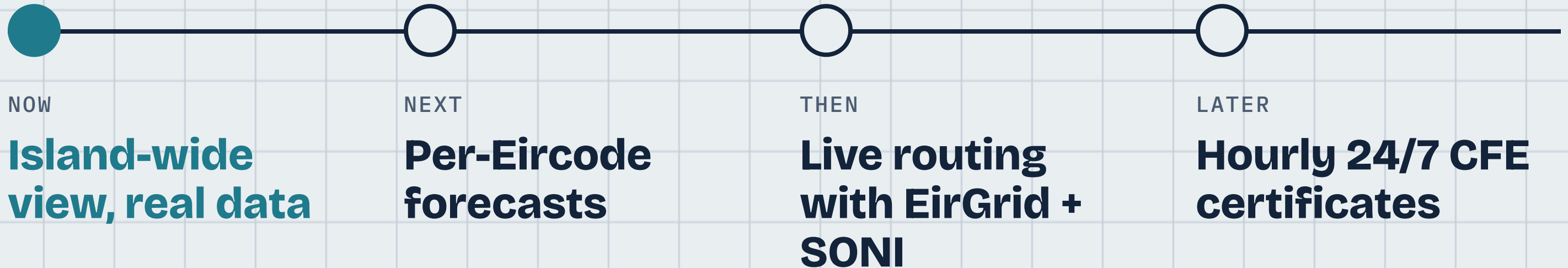
Impact

Technical details

Live demo

What's next

From open data to live routing.



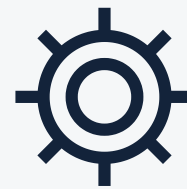
Add-on: flex signals for customers who want more

Three things get us live.



Node-level wind data

EirGrid + SONI



A data-centre pilot

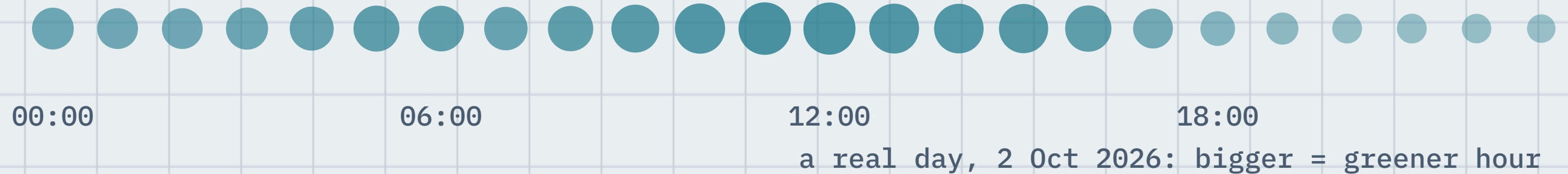
Fingal or Meath



A policy sponsor

forecast-led grid scheduling

Green-hour tokens.

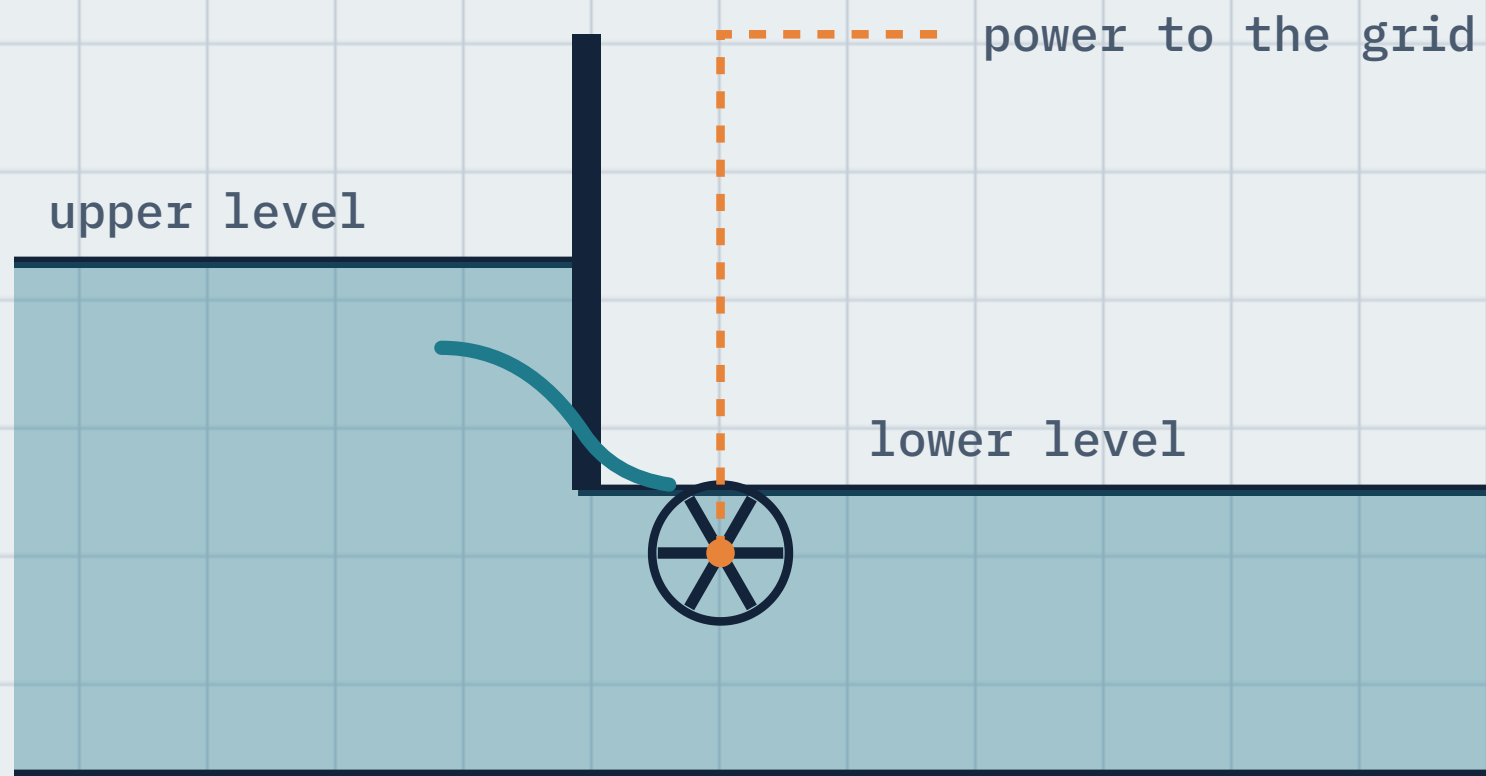


— Fund renewables, earn priority green hours

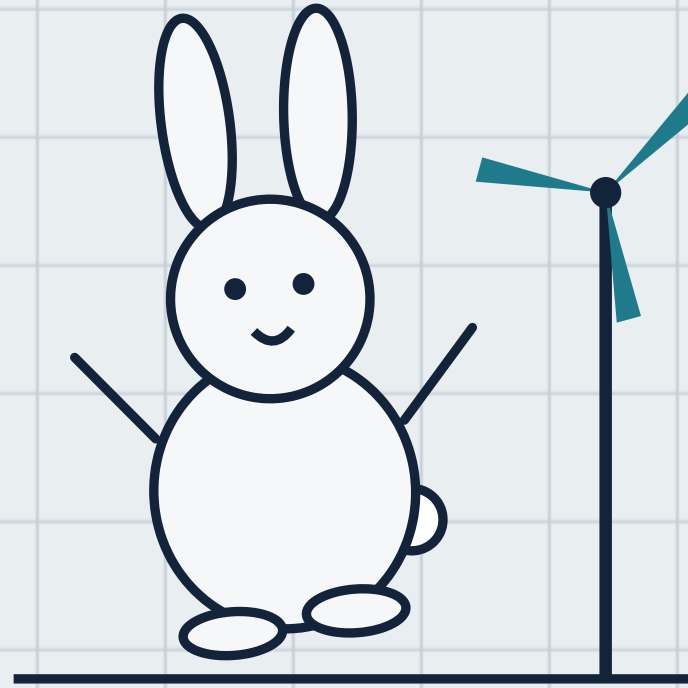
— A cap per company: no cornering the wind

— Every token is an auditable hourly record

Micro-turbines on Ireland's canals.



- Turbines at locks and weirs, where water drops
- Steady output, day and night
- GridSense picks the sites



Thank you.

This bunny's moves are powered by wind
that would otherwise have been switched off.

Questions?